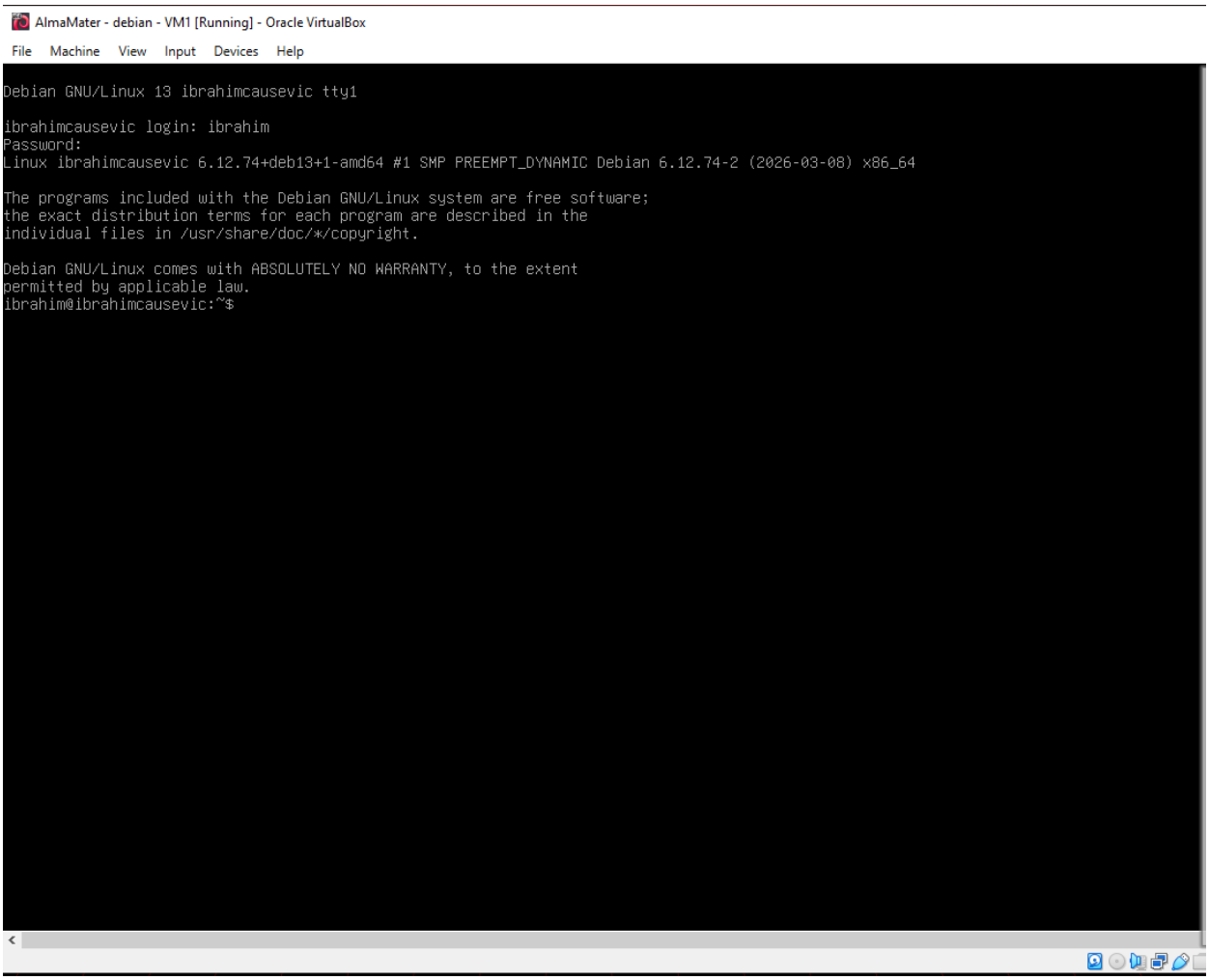
Domača naloga 2 – Ibrahim Čaušević

Virtualni računalnik 1(obstoječi):

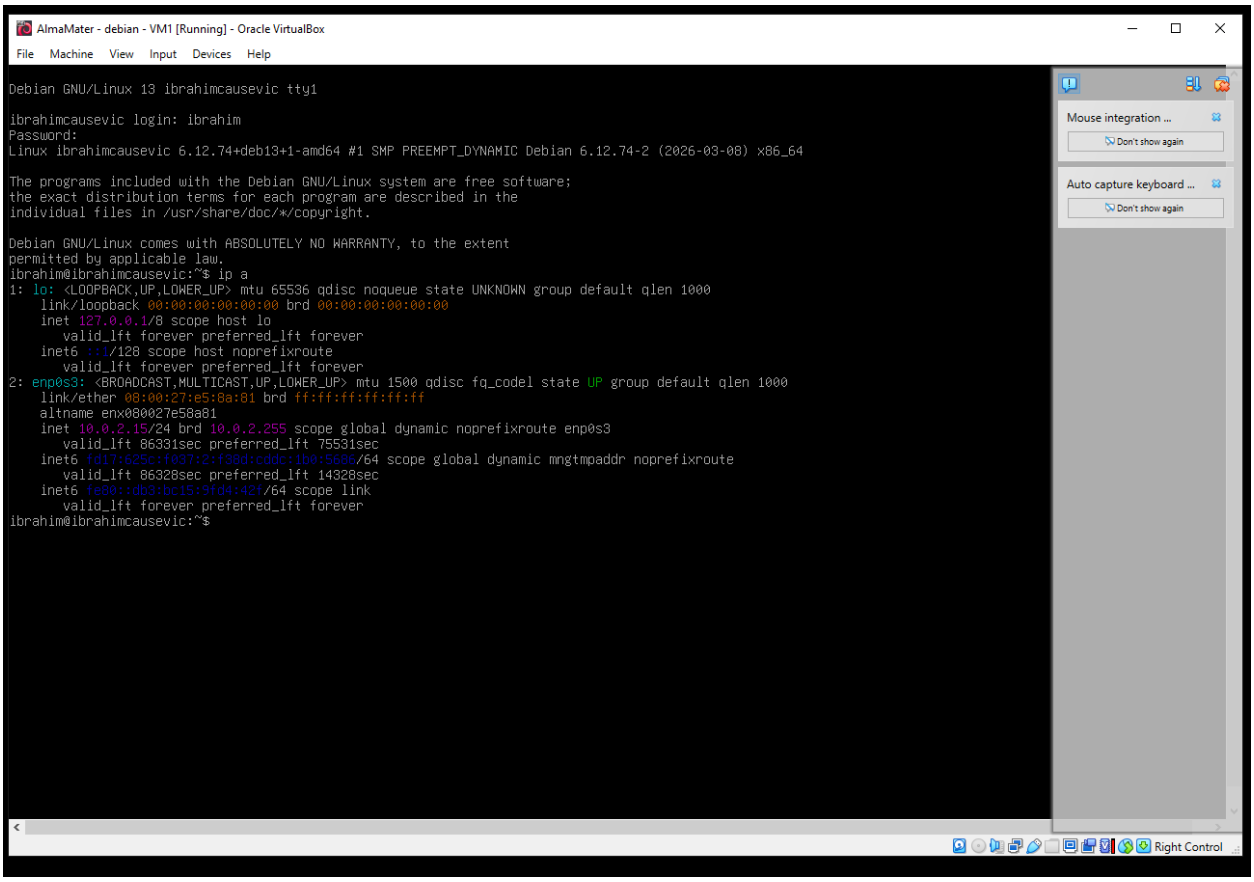
RAM: 500MB

Omrežni vmesnik: NAT

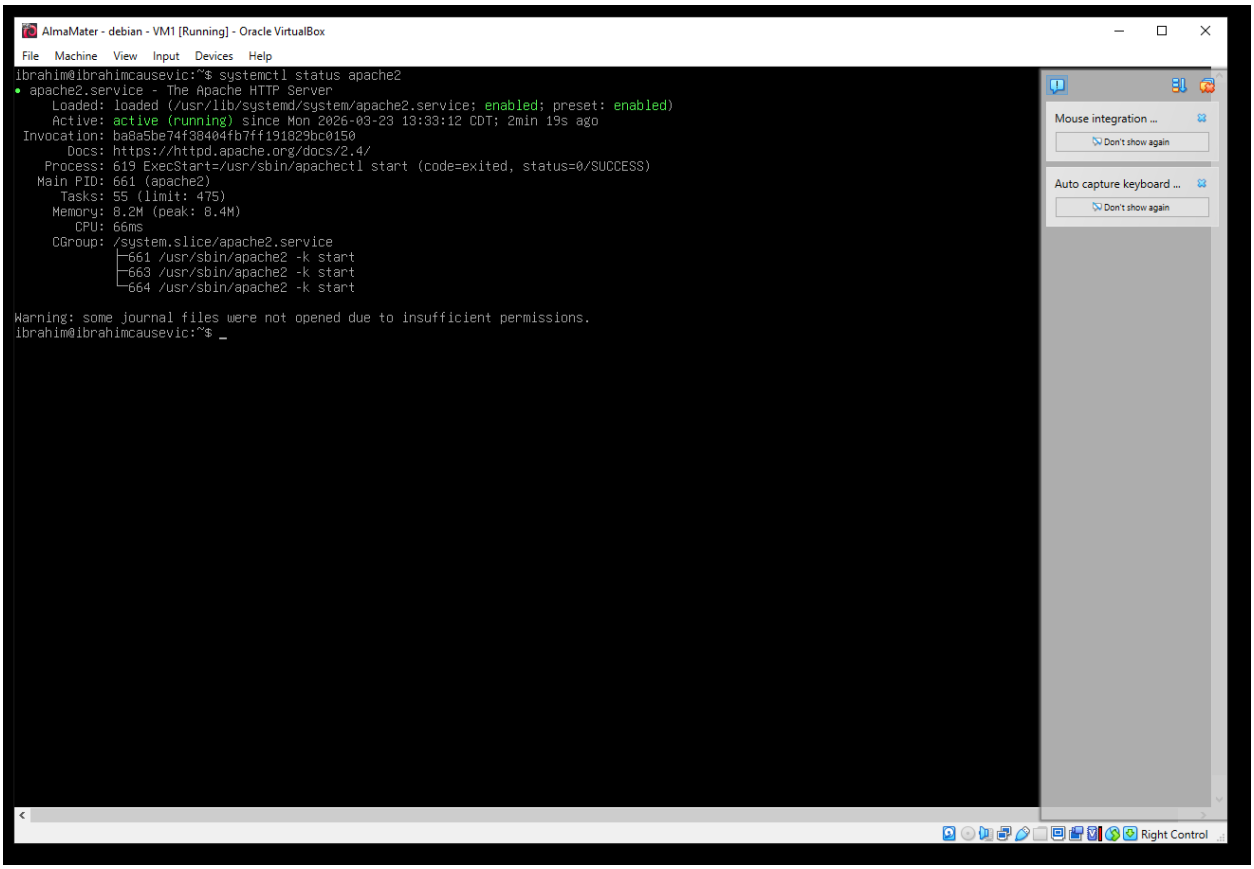
Prijava:



Preverjanje IP naslova virtualnega računalnika 1

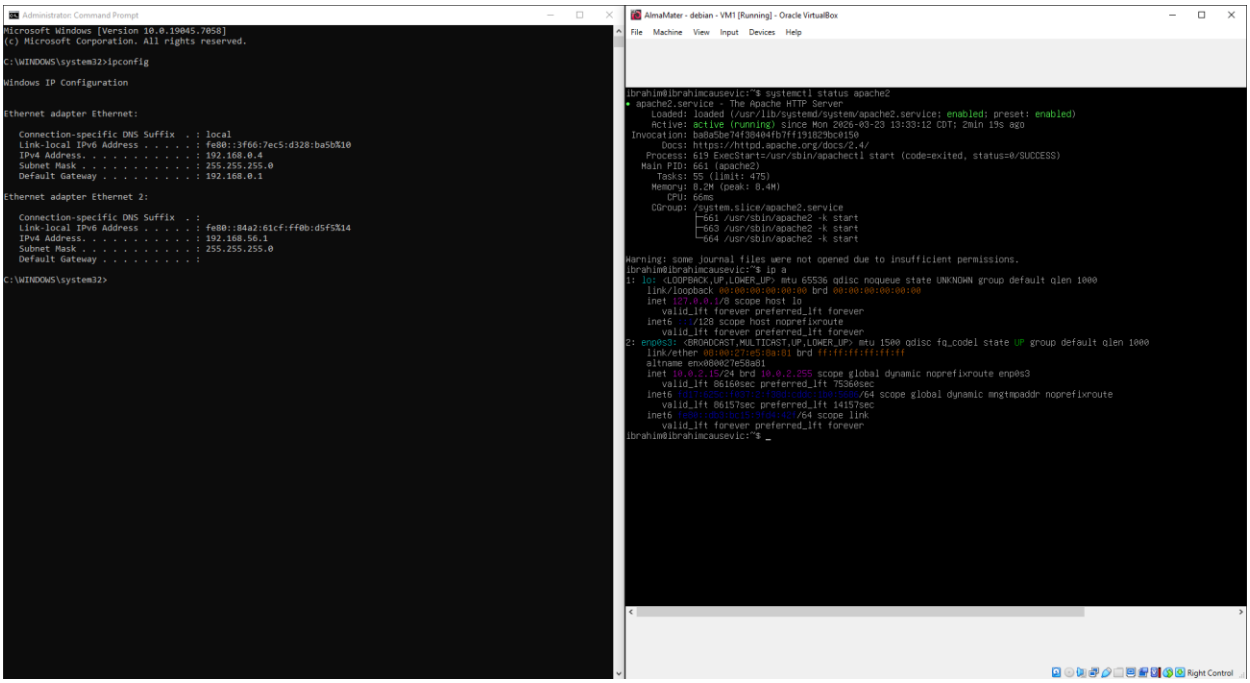


Preverjanje aktivnosti apache2 strežnika: systemctl status apache2



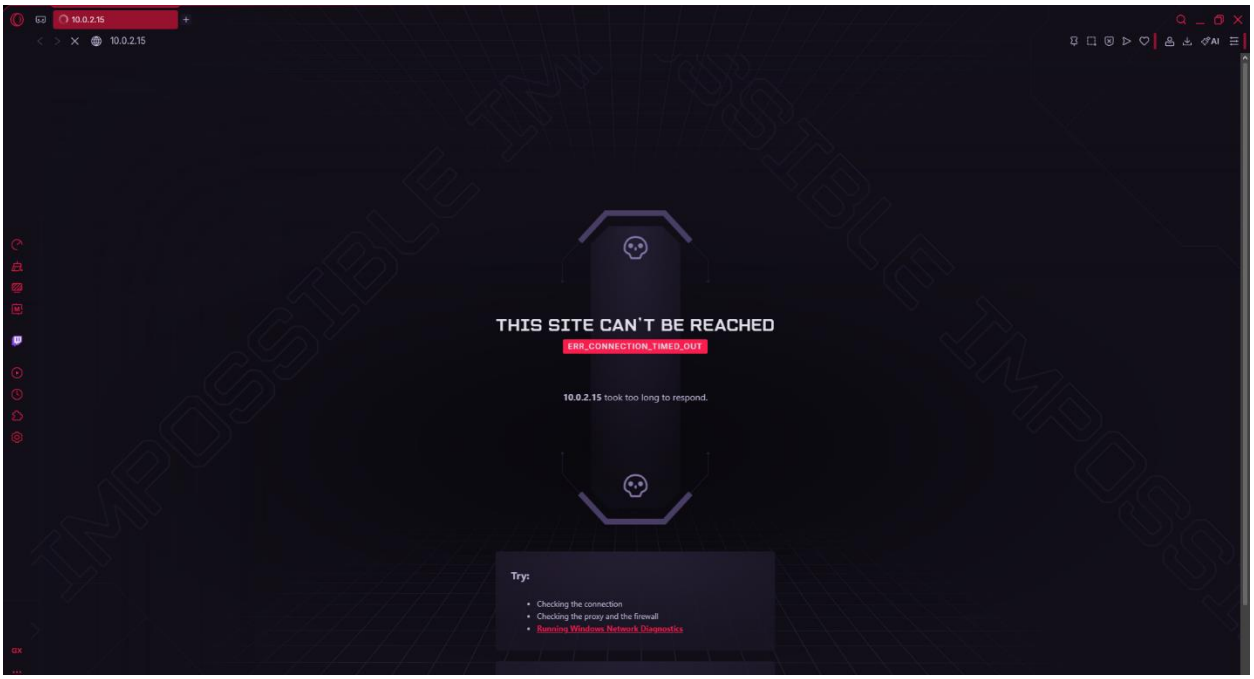
Preverjanje IP naslova fizičnega računalnika in primerjava tega naslova z IP naslovom virtualnega računalnika

Glede na to da je na virtualnem računalniku nastavljen omrežni vmesnik NAT, to pomeni da nista na istem omrežju in zato imata različna IP naslova, oziroma virtualni računalnik uporablja fizični računalnik kot host-a.



Poskus dostopa iz fizičnega računalnika do virtualnega preko SSH(PuTTY) in spletnega brskalnika

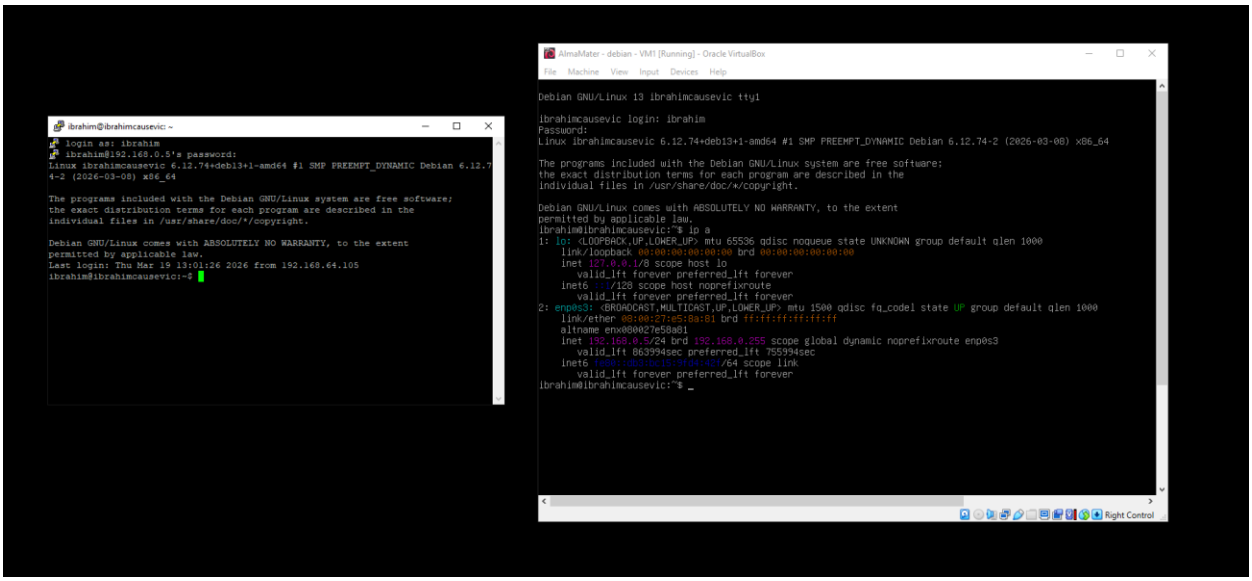
Fizični računalnik ne more direktno dostopati do virtualnega, zato ker je virtualni računalnik na NAT vmesniku, kar pomeni da nista skupaj v omrežju.

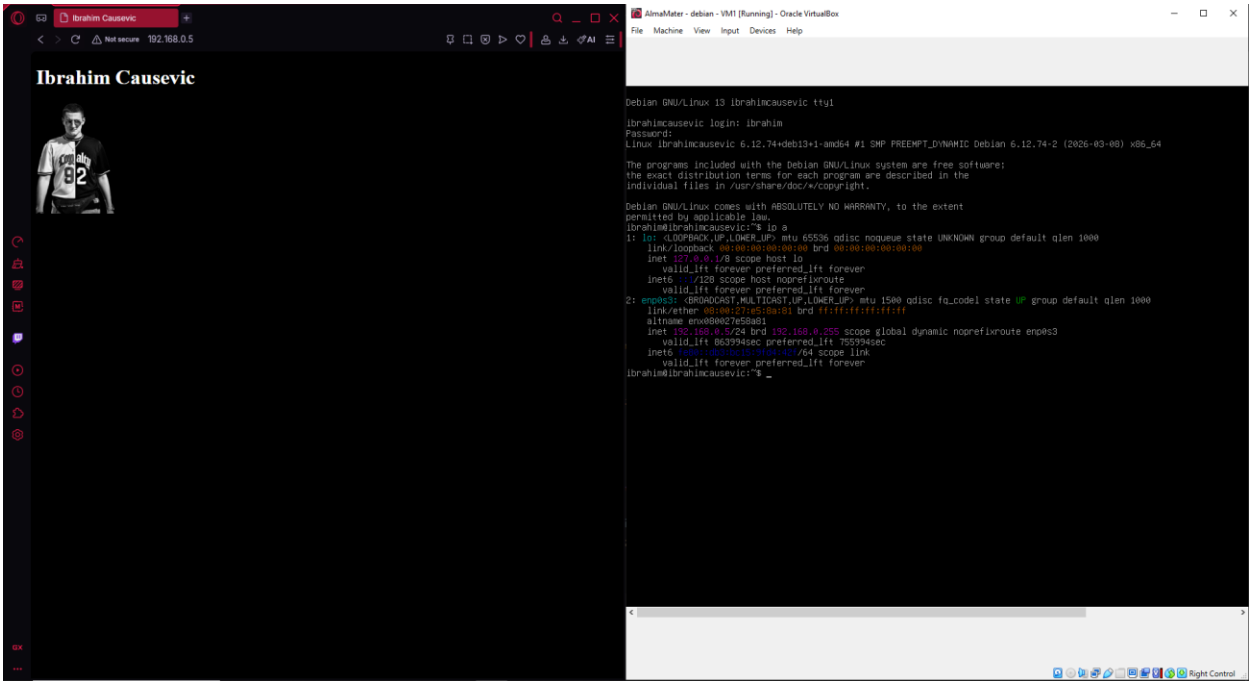


Spremenjen vmesnik virtualnega računalnika na Bridged in preverjanje IP naslova

Zdaj sta fizični in virtualni računalnik na istem omrežju zaradi Bridged vmesnika, zato imata isti naslov, oziroma po IP naslovih se vidi da sta na istem omrežju 192.168.0.x

Dostop prek SSH in brskalnika je možen iz istega razloga, ker sta oba na istem omrežju skupaj.





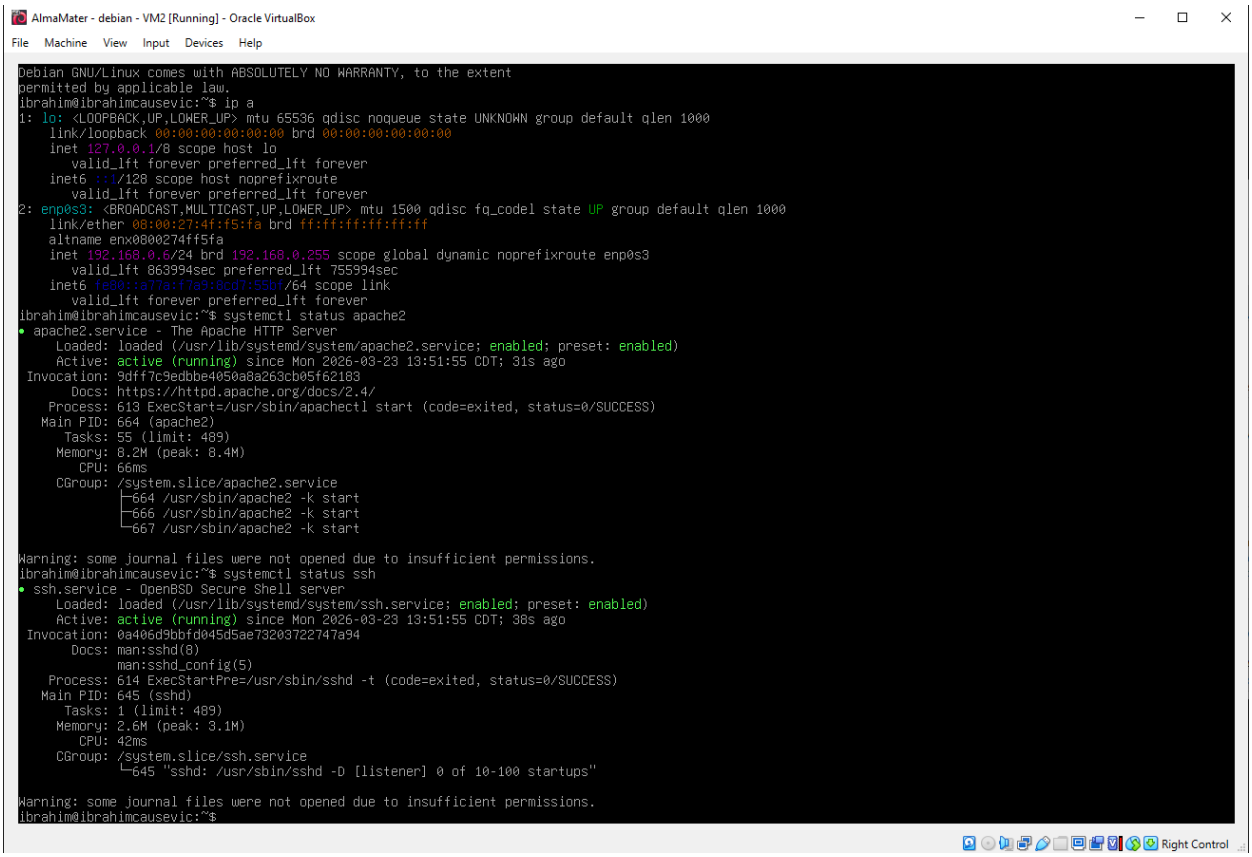
Ustvarjanje novega virtualnega računalnika 2

1 CPU

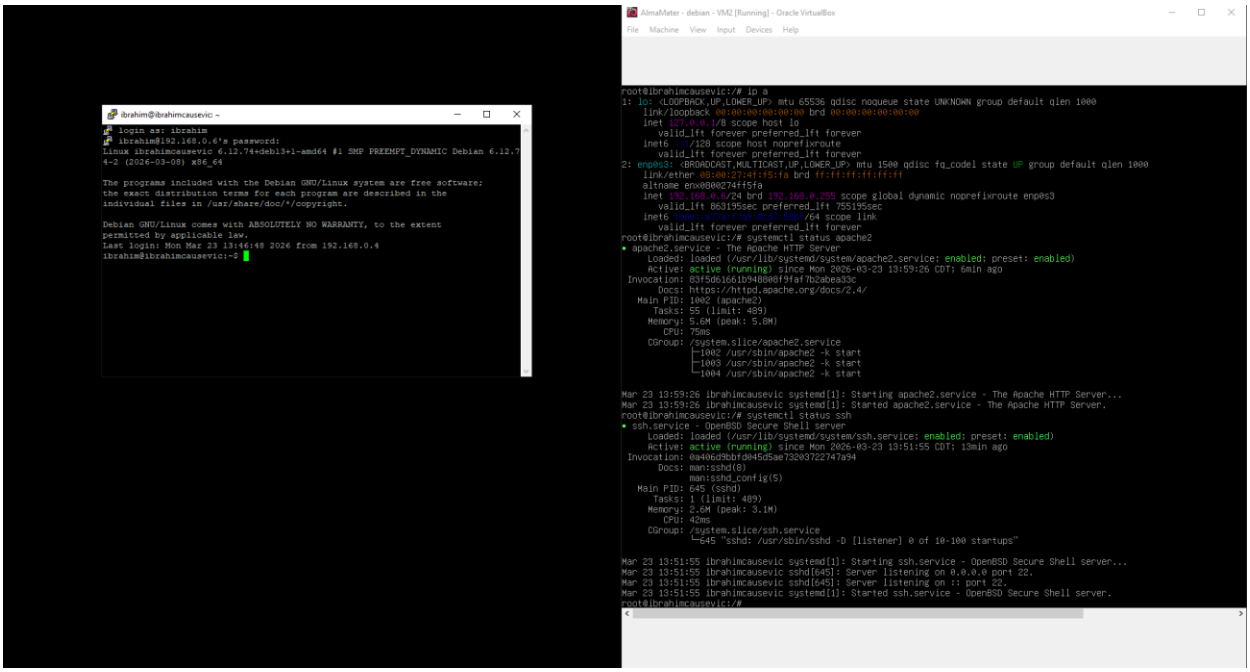
RAM: 512 MB

Omrežni vmesnik: Bridged

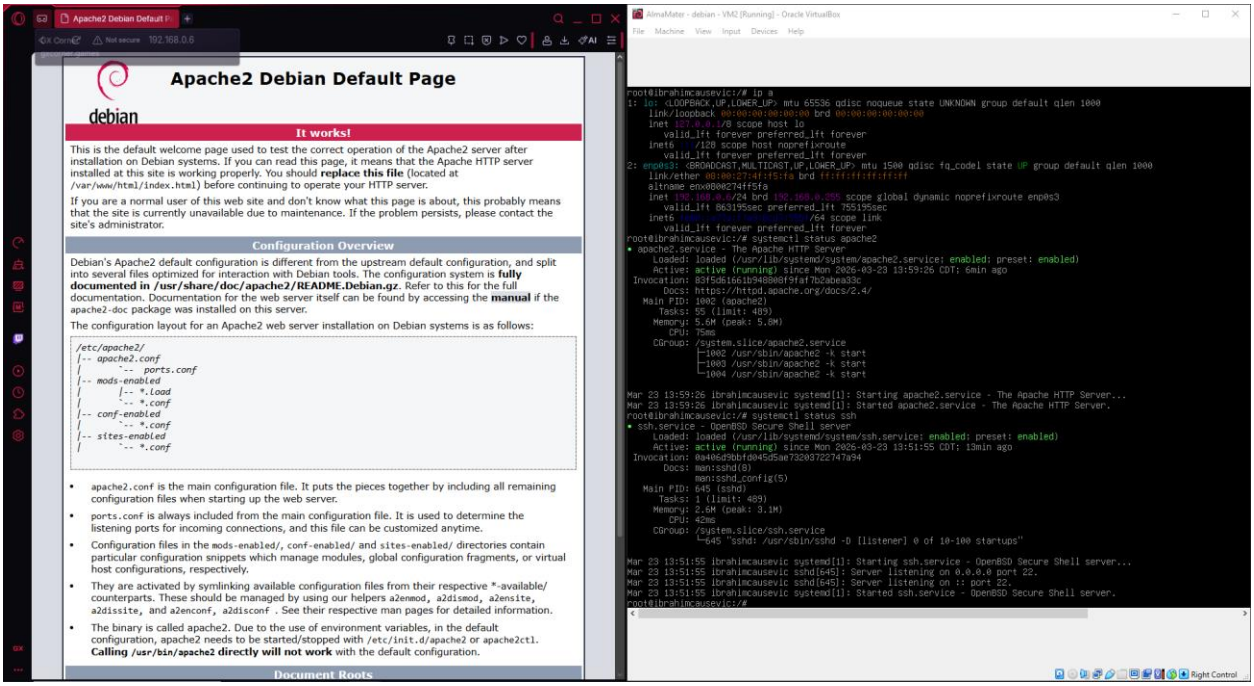
Nameščena SSH in Apache2 strežnika in preverjena aktivnost obeh



Vzpostavljena SSH povezava iz fizičnega računalnika



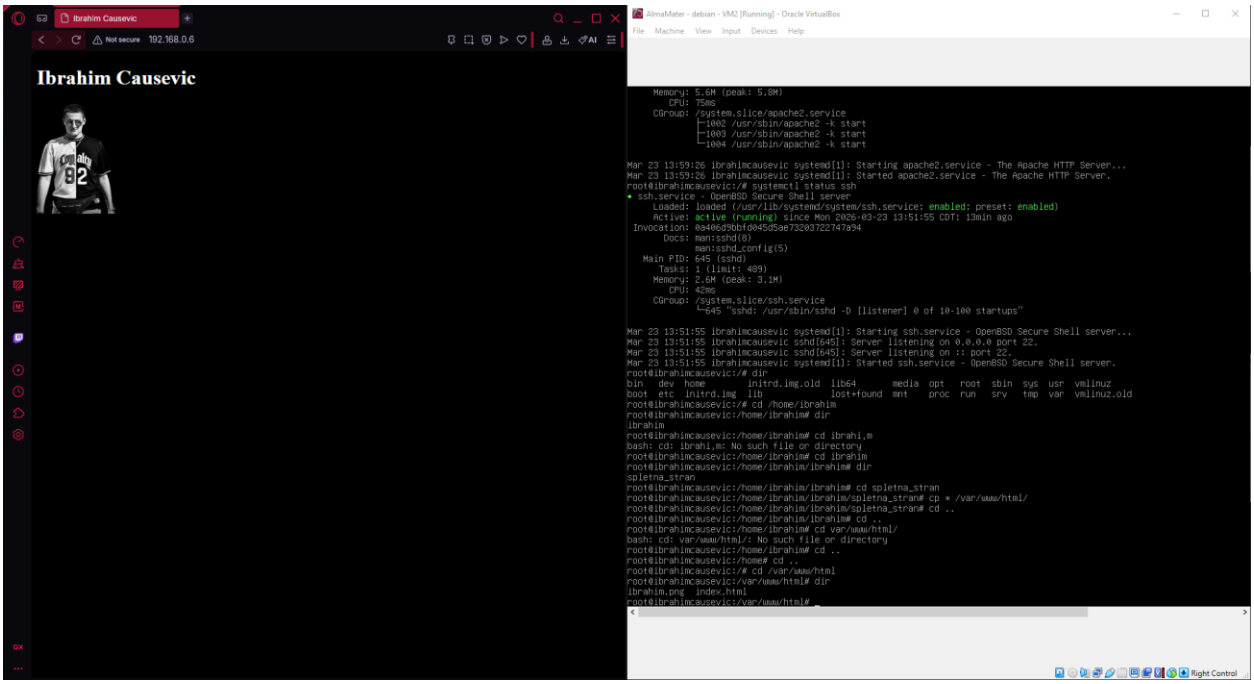
Spletna stran strežnika pokaže, da povezava deluje, ampak je prazna mapa prikaza strežnika. (/var/www/html)



Pošiljanje datotek za spletno stran iz prvega virtualnega računalnika na drugi.(pozabil sem narediti posnetek zaslona)

Uporabljen ukaz na VM1: scp -r /var/www/html/ * ibrahim@192.168.0.6:/home/ibrahim(192.168.0.6 je IP naslov VM2)

Potem pa iz mape /home/ibrahim v mapo /var/www/html za prikaz na strežniku, zato je spletna stran vidna. (ker strežnik ima datoteke za prikazati, prej je mapa bila prazna)

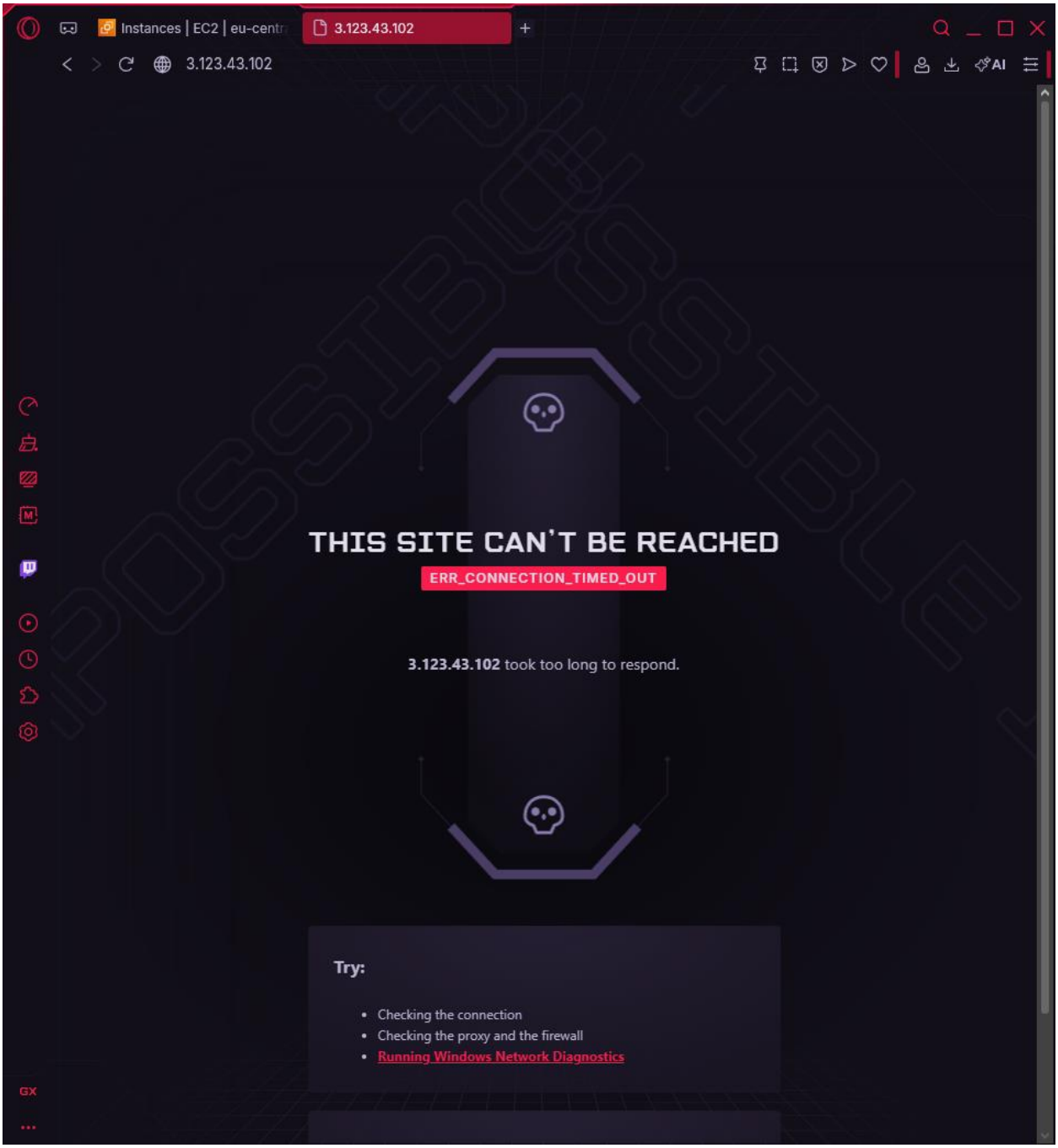


Ustvarjanje nove EC2 instance, tako kot na predavanju. Operacijski sistem Debian z narejenim key pairom: almamaterkey.ppk

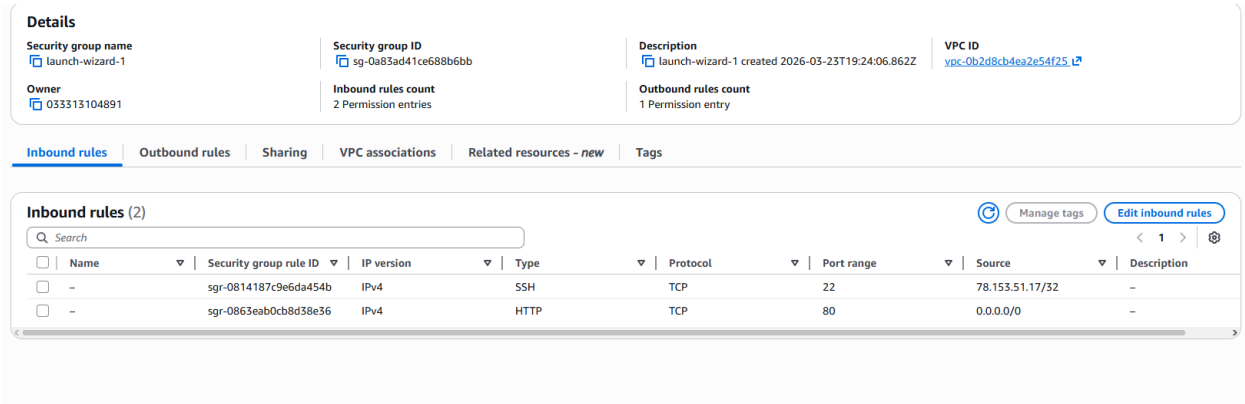
Povezava na strečnik preko SSH in preverjanje če Apache in SSH delujeta.

```
admin@ip-172-31-14-117: ~
Enabling module mime.
Enabling module negotiation.
Enabling module setenvif.
Enabling module filter.
Enabling module deflate.
Enabling module status.
Enabling module reqtimeout.
Enabling conf charset.
Enabling conf localized-error-pages.
Enabling conf other-vhosts-access-log.
Enabling conf security.
Enabling conf serve-cgi-bin.
Enabling site 000-default.
Created symlink '/etc/systemd/system/multi-user.target.wants/apache2.service' → '/usr/lib/systemd/system/apache2.serv
ice'.
Created symlink '/etc/systemd/system/multi-user.target.wants/apache-htcacheclean.service' → '/usr/lib/systemd/system/
apache-htcacheclean.service'.
Processing triggers for man-db (2.13.1-1) ...
Processing triggers for libc-bin (2.41-12+deb13u2) ...
admin@ip-172-31-14-117:~$ systemctl status apache2
● apache2.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
     Active: active (running) since Mon 2026-03-23 19:51:11 UTC; 13s ago
 Invocation: 823342b5c44c405dad12d19d01e7fa85
       Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 1432 (apache2)
      Tasks: 55 (limit: 1097)
     Memory: 5.5M (peak: 5.9M)
        CPU: 47ms
     CGroup: /system.slice/apache2.service
             ├─1432 /usr/sbin/apache2 -k start
             ├─1434 /usr/sbin/apache2 -k start
             └─1435 /usr/sbin/apache2 -k start

Mar 23 19:51:11 ip-172-31-14-117 systemd[1]: Starting apache2.service - The Apache HTTP Server...
Mar 23 19:51:11 ip-172-31-14-117 apachectl[1431]: AH00558: apache2: Could not reliably determine the server's fully >
Mar 23 19:51:11 ip-172-31-14-117 systemd[1]: Started apache2.service - The Apache HTTP Server.
admin@ip-172-31-14-117:~$ systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
     Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
     Active: active (running) since Mon 2026-03-23 19:36:58 UTC; 15min ago
 Invocation: 4f64552b962f4e4cb6582a78f61f1128
       Docs: man:sshd(8)
             man:sshd_config(5)
   Main PID: 722 (sshd)
      Tasks: 1 (limit: 1097)
     Memory: 7.8M (peak: 9.6M)
        CPU: 152ms
     CGroup: /system.slice/ssh.service
             └─722 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Mar 23 19:36:58 ip-172-31-14-117 sshd[722]: Server listening on :: port 22.
Mar 23 19:36:58 ip-172-31-14-117 systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
Mar 23 19:38:32 ip-172-31-14-117 sshd-session[784]: Invalid user ibrahim from 78.153.51.17 port 65457
Mar 23 19:38:41 ip-172-31-14-117 sshd-session[784]: error: Received disconnect from 78.153.51.17 port 65457:14: No s>
Mar 23 19:38:41 ip-172-31-14-117 sshd-session[784]: Disconnected from invalid user ibrahim 78.153.51.17 port 65457 [>
Mar 23 19:40:15 ip-172-31-14-117 sshd-session[786]: Invalid user ibrahimc from 78.153.51.17 port 65467
Mar 23 19:40:16 ip-172-31-14-117 sshd-session[786]: error: Received disconnect from 78.153.51.17 port 65467:14: No s>
Mar 23 19:40:16 ip-172-31-14-117 sshd-session[786]: Disconnected from invalid user ibrahimc 78.153.51.17 port 65467 >
Mar 23 19:40:40 ip-172-31-14-117 sshd-session[788]: Accepted publickey for admin from 78.153.51.17 port 65504 ssh2: >
Mar 23 19:40:40 ip-172-31-14-117 sshd-session[788]: pam_unix(sshd:session): session opened for user admin(uid=1000) >
admin@ip-172-31-14-117:~$
```

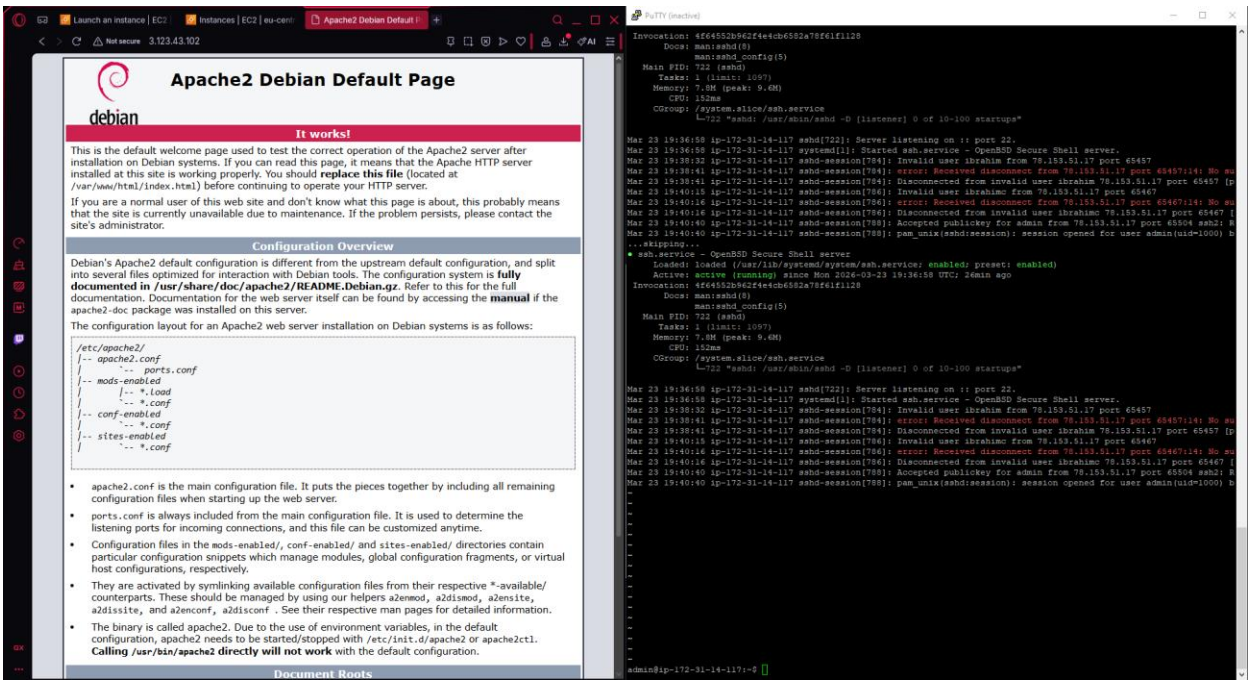
Poskus dostopa do strežnika preko brskalnika



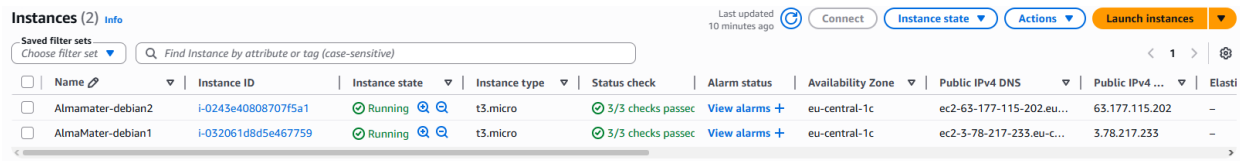
Dodan inbound rule za HTTP protokol, da se omogoča dostop.



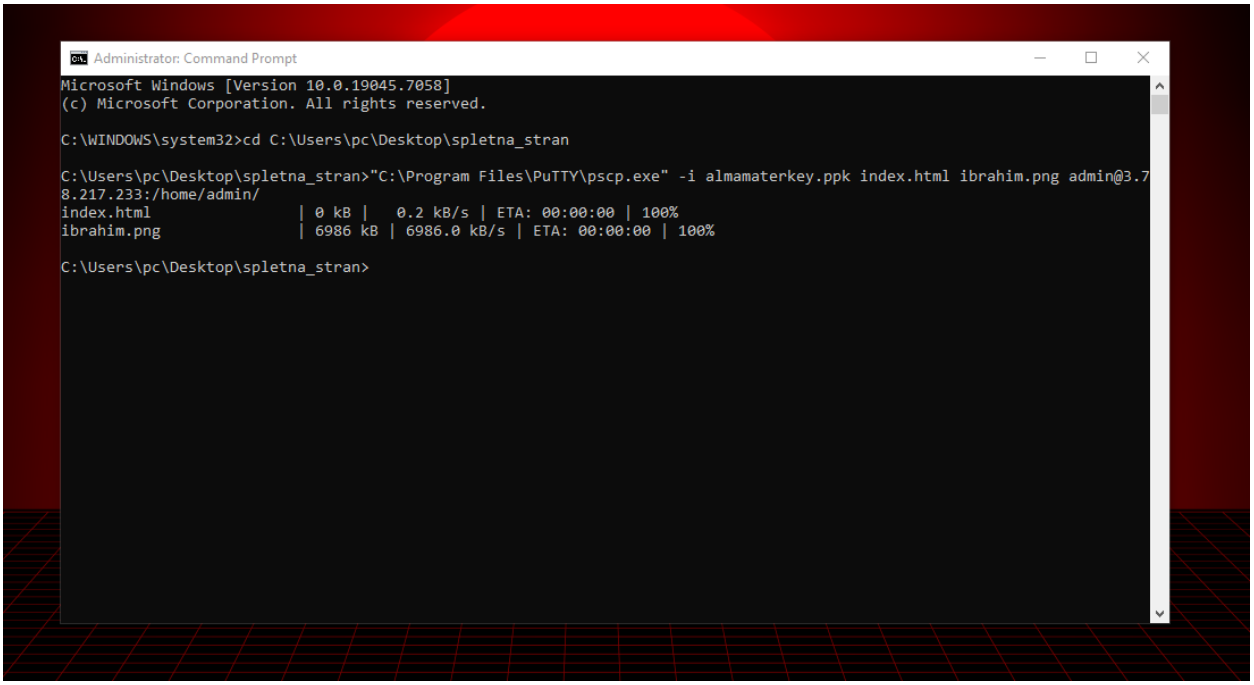
Prikaz default strani na strežniku, spet zaradi prazne mape strežnika.



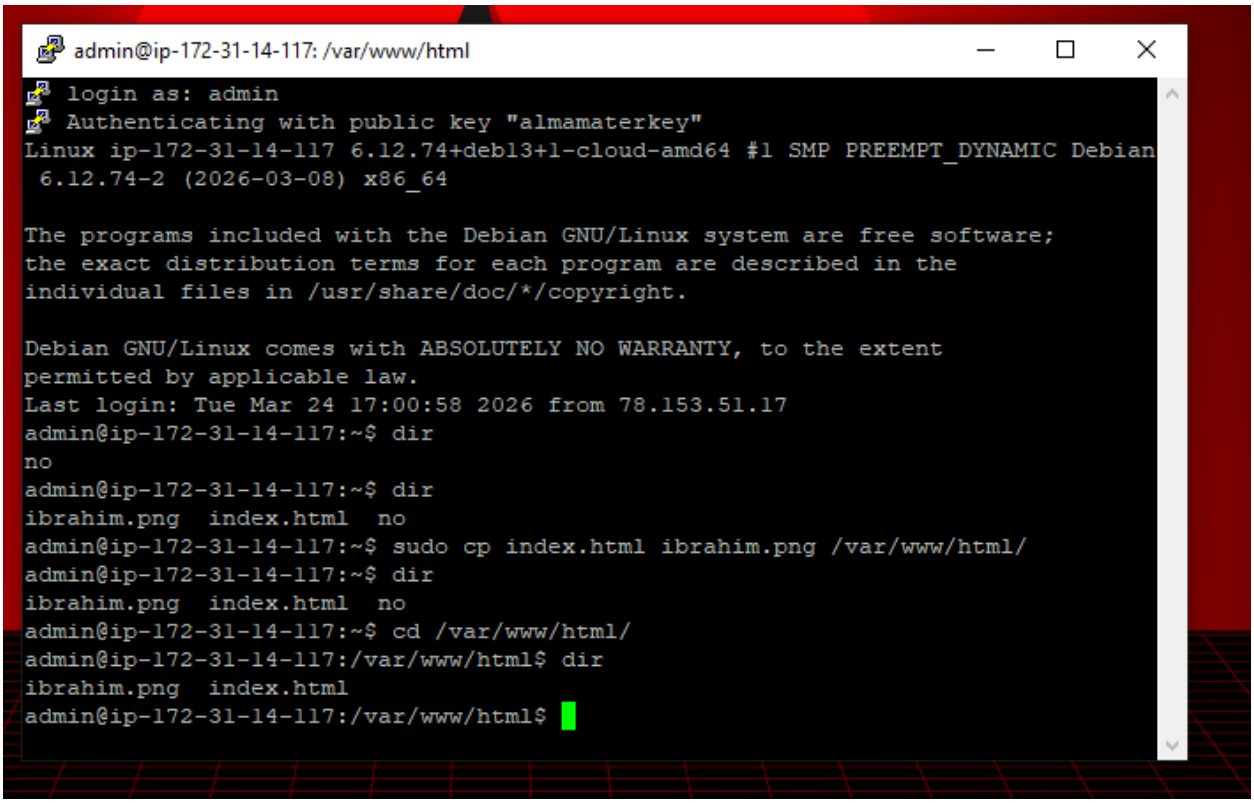
Vmes izdelana druga instanca na AWS



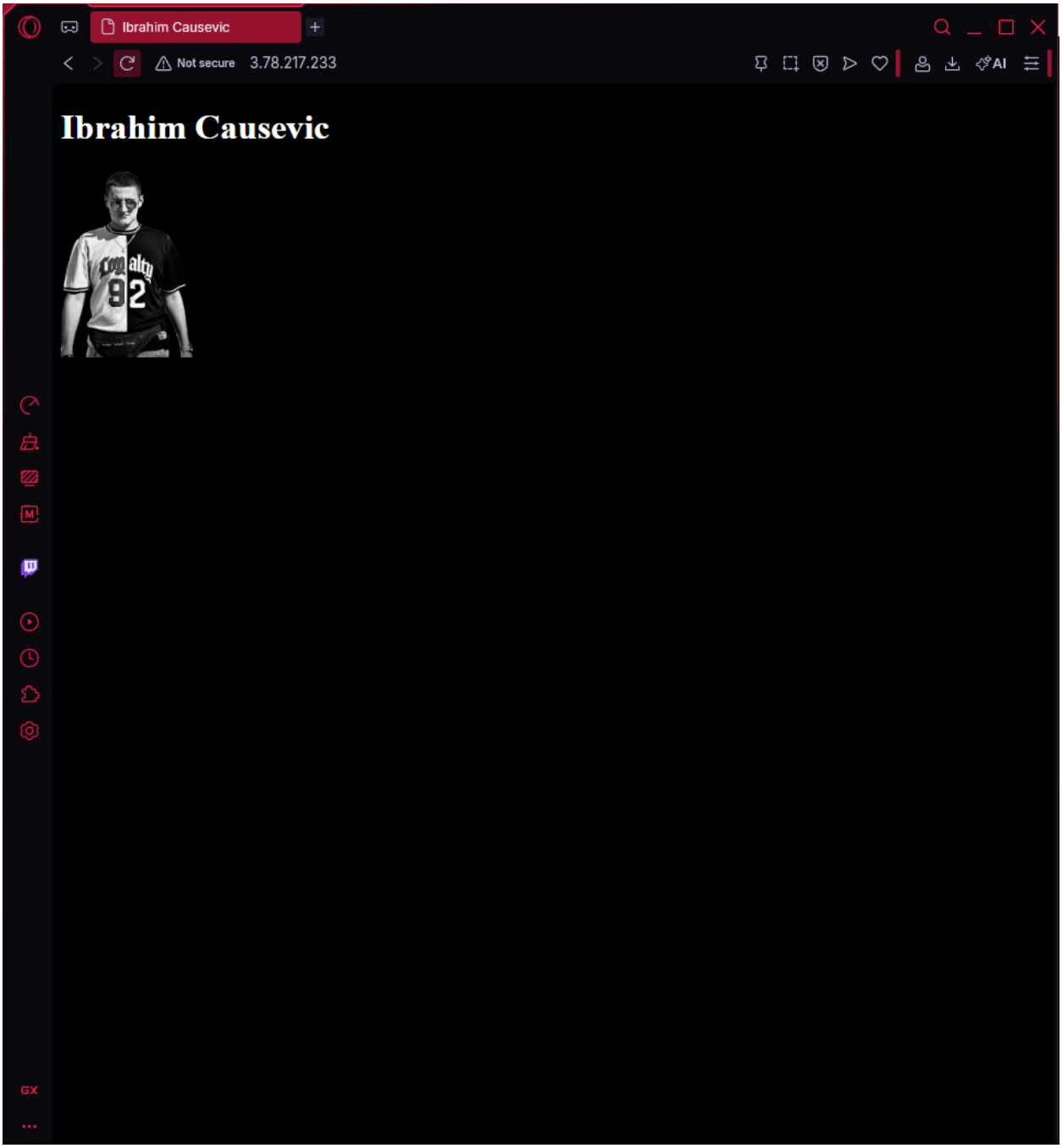
Prenašanje datotek za spletno stran iz fizičnega računalnika na prvo instanco.



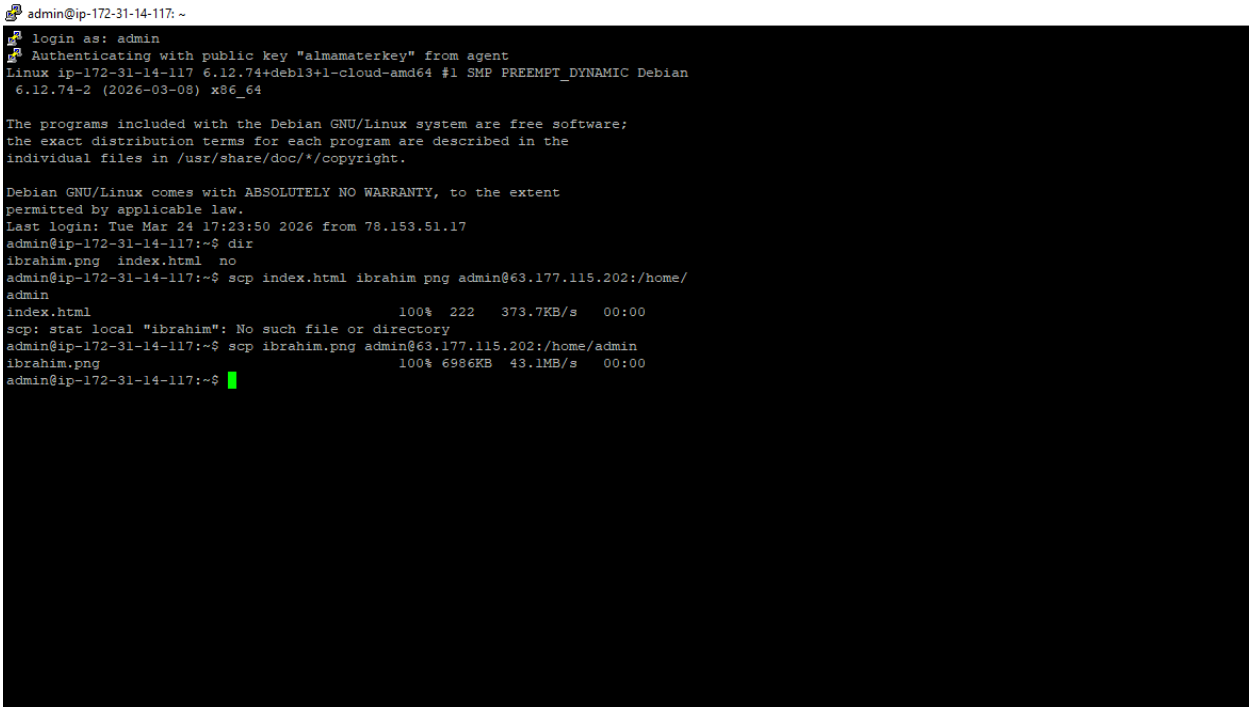
Prenos datotek na v mapo /var/www/html za prikaz na strežniku



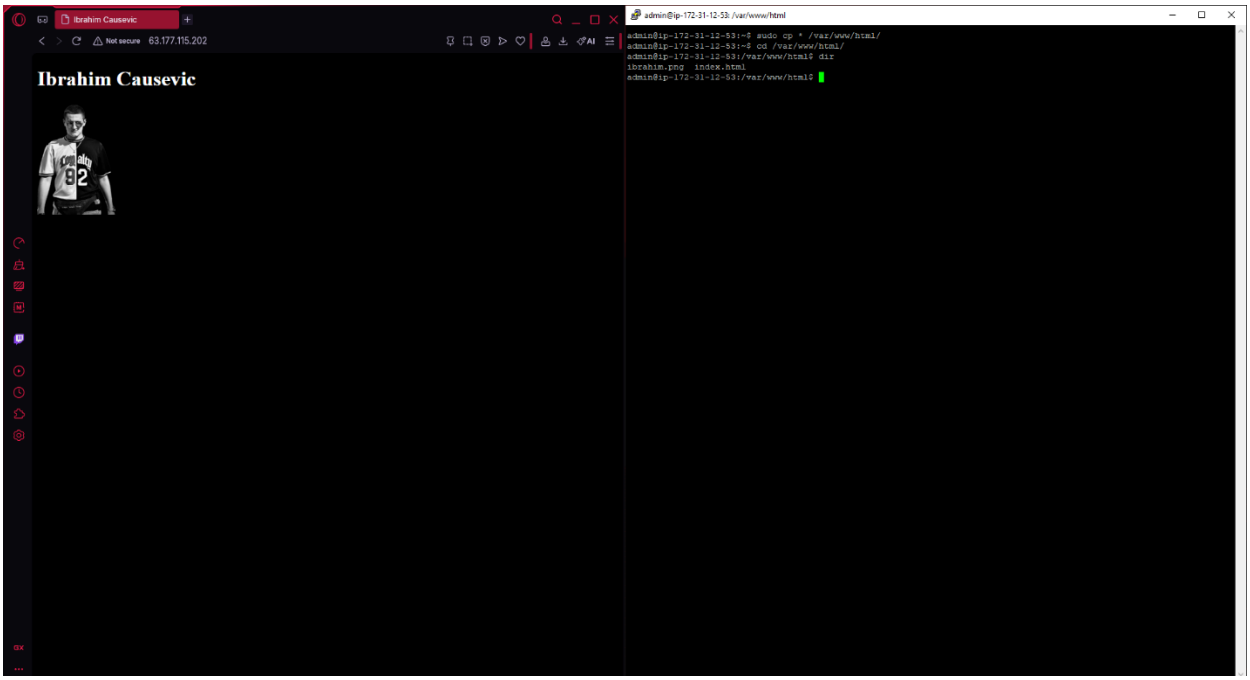
Prikaz spletne strani v brskalniku



Prenašanje spletne datoteke iz prve instance na drugo, key pair sem dodal v aplikacijo Pageant, ki je child aplikacija od PuTTY-a, da lahko uporabi key pair za dostop do druge instance.



Premik datotek v mapo /var/www/html/ in prikaz strani v brskalniku



- Kakšna je razlika med NAT in Bridged omrežjem?

V NAT virtualni računalnik in fizični računalnik nista v istem omrežju, zato virtualni računalnik uporablja fizičnega kot hosta za dostop do interneta, virtualni pa je v tem primeru guest.

V Bridged pa sta v istem omrežju, fizični računalnik mu ni host ampak samo še ena naprava. Virtualna mašina je vidljiva v lokalnem omrežju.

- Zakaj v NAT načinu dostop pogosto ne deluje?

Zato ker virtualna mašina ni v lokalnem omrežju in host ne more dostopati do virtualnega računalnika preko SSH ali brskalnika, ker virtualni računalnik ima interni IP naslov v NAT.

- Kakšna je razlika med javnim in zasebnim IP naslovom?

Razlika med javnim in zasebnim IP naslovom je to, da je javni IP naslov dosegljiv prek interneta iz drugih omrežij, zasebni pa ne, zasebni je dosegljiv lokalno.

- Kakšna je vloga SSH?

SSH je protokol ki omogoča varno upravljanje računalnikov in strežnikov na daljavo.

- Kakšna je vloga Apache strežnika?

Vloga apache strežnika je da varno posreduje spletne vsebine in obdeluje zahtevke prek http protokolov.

- Kaj omogoča Security Group v AWS?

Security group v AWS nam omogoča da upravljamo dostope do instance. Odpiramo različne porte, različne protokole za različne IP naslove.